

FinRAG Project — Progress Report

Phase: Data Pipeline, Graph Construction & Initial Retrieval Tiers

1. Dataset Acquisition and Filtering

- Downloaded the full **FinReflectKG** dataset from Hugging Face (`domyn/FinReflectKG`) — 17.5 million entity-relation triplets extracted from S&P 500 10-K filings (2014–2024).
- Filtered the dataset to Apple (AAPL) and Microsoft (MSFT) only, resulting in a working subset of **51,927 triplets**:
 - MSFT: 30,891 triplets
 - AAPL: 21,036 triplets
 - Coverage spans 2014–2024 for both companies, with no year gaps.
- Confirmed dataset schema: `triplet_id, entity, entity_type, relationship, target, target_type, start_date, end_date, extraction_type, ticker, year, source_file, page_id, chunk_id, chunk_text, triplet_length, chunk_text_length, has_context`.
- Saved the filtered subset locally as a CSV for repeated use without re-downloading or re-filtering.

2. Knowledge Graph Construction (Neo4j AuraDB)

- Set up a Neo4j AuraDB (cloud, free tier) instance for the project.
- Designed a graph schema:
 - **Nodes**: `Entity`, labeled with `entity_type` (e.g., `ORG`, `PERSON`, `FINMETRIC`, `RISKFACTOR`, `ORGREG`, `FINMARKET`, etc.)
 - **Edges**: `RELATION`, carrying the relationship type, dates, year, source filing, chunk ID, and the original source text (`chunk_text`) for provenance.
- Created an index (`entity_name_ticker`) on entity name + ticker to speed up `MERGE` operations during loading.
- Loaded all 51,927 triplets into AuraDB using a batched (500 rows/batch), `MERGE` -based Cypher loading script — avoids duplicate nodes/edges on re-runs.
- **Verified load**: 15,663 unique nodes, 51,927 relationships — relationship count matches the source CSV exactly, confirming a clean load with no data loss.

3. Graph Pattern Exploration

- Queried the graph to discover the most frequent 2-hop and 3-hop relationship patterns actually present in the AAPL/MSFT subset, rather than assuming which patterns exist.

- Key learning: naive frequency counting (`count( * )`) produced impossible numbers (in the hundreds of thousands) due to combinatorial explosion through high-degree "hub" nodes (e.g., the AAPL/MSFT organization nodes, which connect to thousands of edges). Fixed by counting **distinct entity-triples** instead of raw path matches.
- Found dominant 2-hop patterns are largely governance/ownership-related (e.g., `PERSON - [has_stake_in]→ ORG - [discloses]→ FIN_METRIC`), while more analytically useful patterns (e.g., risk and market-impact chains) are rarer but genuinely present.
- Noted a **data quality observation**: some entity nodes (e.g., generic metric names like "net income") are reused across multiple years without a distinguishing year attached to the node itself, which can cause path-based queries to stitch together facts from unrelated years. This is a limitation inherited from the underlying KG construction, not from our pipeline, and is worth citing as an observed limitation in the final report.

4. Sample Question Set (Manual Mini-Benchmark)

Since the full FinReflectKG-MultiHop benchmark question set was not accessible during this phase, a small manually-verified sample set was built directly from confirmed graph patterns:

- **Q1** (2-hop, regulatory): *"Which regulatory bodies are named in connection with Apple's disclosed net income, and what litigation prompted this?"* Ground truth (from graph): U.S. Department of Justice, SEC — tied to the Apple eBooks Antitrust Litigation (filed April 2012).
- **Q2** (2-hop, market risk): *"What financial market conditions does Apple disclose as negatively impacting its financial metrics?"* Ground truth (from graph): liquidity, credit market, fixed income market — from the Risk Factors section.

This mini-benchmark is being used to test each retrieval tier before scaling to the full benchmark.

5. Symbolic Computation Tool

- Built a tool that fetches numeric facts from the graph across years for a given metric and computes derived values (e.g., year-over-year growth), with results traceable back to the exact `triplet_id` and source filing used.
- **First attempt** revealed a numeric extraction bug: the initial regex-based number extractor grabbed the first number-like pattern in a chunk (often a date or header) rather than the actual metric value, producing a nonsensical growth rate.
- **Fixed** by searching for numbers specifically following the metric's label (e.g., "Net sales") rather than anywhere in the chunk — this produced correct-looking numbers (\$182,795M and \$155,041M).
- **Second issue discovered**: even with correct-looking numbers, the computed growth rate was still wrong, because 10-K filings contain multiple tables labeled "Net sales" at different levels of granularity (consolidated total vs. segment/product/region breakdowns). The tool had pulled a segment-level figure rather than the consolidated total for one of the two years, producing a misleading result.
- **Decision**: rather than building a full table-disambiguation system (a significant additional scope), this is documented as a known limitation. The tool's core pipeline — fetch, extract, compute, and trace to

source — works correctly end-to-end; achieving perfectly accurate figure extraction in ambiguous multi-table filings is identified as a valuable direction for further refinement.

6. Ablation Tiers — Progress So Far

Tier 0 — No Retrieval (Baseline)

- Asked an LLM (Llama 3.3 70B via Groq) the sample questions directly, with no context or grounding.

Q1 — Tier 0 Answer:

Apple's net income is not directly regulated by any specific regulatory body as it's a financial metric that companies disclose voluntarily. However, Apple, like many multinational corporations, is subject to various regulations and oversight from several regulatory bodies in different countries related to their business operations, tax practices, data privacy, and consumer protection.

The litigation that often comes up in connection with Apple's net income primarily revolves around tax-related issues. For instance: 1. European Commission: In 2016, the European Commission ordered Ireland to recover up to €13 billion (\$15 billion) in back taxes from Apple due to state aid rules being broken by granting undue tax benefits to Apple for over two decades. 2. United States: The Senate Permanent Subcommittee on Investigations (PSI) launched an investigation into Apple's international tax strategies in 2013... 3. India: In 2020, the Indian Tax Department imposed a tax demand of around \$650 million on Apple... 4. Other countries like Australia, France, and the UK have also been involved in tax disputes with Apple over the years.

Q2 — Tier 0 Answer:

Apple discloses various financial market conditions that may affect its business operations and financial metrics. These can include: 1. Economic conditions: Slower economic growth or recession in key markets... 2. Currency fluctuations: Changes in foreign exchange rates can impact Apple's reported earnings... 3. Interest rates: Higher interest rates can increase Apple's cost of debt... 4. Competitive factors: Intense competition from other technology companies... 5. Regulatory and policy changes: Changes in regulations or policies related to data privacy, digital services taxes... 6. Supplier constraints: Supply chain disruptions due to factors such as geopolitical tensions... 7. Technology shifts: Changes in consumer preferences toward emerging technologies...

- **Result:** The model hallucinated plausible but incorrect specifics.
 - Q1: Invented unrelated litigation (EU state-aid tax ruling, Senate PSI investigation, India tax dispute) — never mentioned the actual eBooks antitrust case.
 - Q2: Gave generic, non-filing-specific categories (economic conditions, currency fluctuations, interest rates) rather than Apple's actual disclosed language (liquidity, credit market, fixed income market).
- **Conclusion:** Confirms the core problem motivating this project — an ungrounded LLM produces confident-sounding but factually wrong or overly generic financial answers.

Tier 1 — Vector RAG (Retrieval, No Graph)

- Embedded all unique `chunk_text` passages for AAPL using a sentence embedding model, retrieved top-k similar chunks per question, and passed them to the LLM.

First attempt (`all-MiniLM-L6-v2`):

Q1 — Answer:

The context doesn't contain the answer. There is no mention of regulatory bodies or litigation related to Apple's disclosed net income. The context only appears to provide headers for various financial statements and some notes related to revenue categories, but it does not discuss net income, regulatory bodies, or litigation.

Q2 — Answer:

The context does not contain the answer. The provided context appears to be excerpts from Apple Inc.'s financial reports and auditor's opinions, but it does not discuss specific financial market conditions that negatively impact Apple's financial metrics.

- Retrieval failed for both questions — the model matched generic financial-document "vibes" (headers, boilerplate: Consolidated Statements of Cash Flows, auditor opinion letters) rather than the specific relevant passages, and the LLM correctly reported it could not find an answer.

Second attempt (`BAAI/bge-small-en-v1.5` , with query-instruction prefix):

Q1 — Answer (correct):

The regulatory bodies mentioned in the context of Apple's disclosed net income are the U.S. Securities and Exchange Commission (SEC) and the U.S. Department of Justice. The litigation that prompted this disclosure is the Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.), which was filed by the U.S. Department of Justice on April 11, 2012, against Apple and five major book publishers. The lawsuit alleges an unreasonable restraint of interstate trade and commerce in violation of Section 1 of the Sherman Act.

Q2 — Answer (still incorrect):

The text does not contain the answer to the question regarding the specific financial market conditions that Apple discloses as negatively impacting its financial metrics.

- **Q1: Succeeded.** Retrieved the exact eBooks Antitrust Litigation passage; LLM correctly answered with SEC and DOJ, matching ground truth.

- **Q2: Still failed.** The correct passage (liquidity/credit market risk factors) was not retrieved even in the top 8–20 results — semantic similarity search ranked generic accounting-policy and auditor-opinion text higher, since financial/legal boilerplate broadly "sounds similar" across a 10-K.
- **Conclusion:** Pure vector search is a real improvement over no retrieval, but has a measurable, specific weakness — it can miss precise structured facts (like typed risk categories) when the surrounding document is dense with similar-sounding legal/financial language. This directly motivates the next tier.

Tier 2 — Graph-Enriched RAG (Vector Search + Direct Graph Traversal)

- Combined vector search with a direct, targeted graph query: for each question, relevant relationship types (e.g., `regulates`, `negatively_impacts`, `impacted_by`) were queried directly against the graph, guaranteeing retrieval of structurally relevant facts regardless of semantic phrasing.

Q1 — Graph facts retrieved:

```
u.s. department of justice (ORG_REG) --regulates--> AAPL
u.s. district court for the southern district of new york (ORG_REG) --regulates--> AAPL
sec (ORG_REG) --regulates--> AAPL    [x7]
ireland (GPE) --regulates--> AAPL
```

Q1 — Answer (correct):

The regulatory bodies named in connection with Apple are: 1. U.S. Department of Justice (ORGREG) 2. U.S. District Court for the Southern District of New York (ORGREG) 3. SEC (ORG_REG)

The litigation that prompted this is the Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.), which was filed by the U.S. Department of Justice on April 11, 2012, alleging an unreasonable restraint of interstate trade and commerce in violation of §1 of the Sherman Act.

Q2 — Graph facts retrieved:

```
production ramp (FIN_METRIC) --impacted_by--> AAPL
channel partner (COMP) --negatively_impacts--> AAPL
price reduction (CONCEPT) --negatively_impacts--> AAPL    [x2]
stock price (FIN_METRIC) --impacted_by--> AAPL
liquidity (FIN_MARKET) --negatively_impacts--> AAPL
outsource partner (LOGISTICS) --negatively_impacts--> AAPL
supplier (COMP) --negatively_impacts--> AAPL
third-party developer (COMP) --negatively_impacts--> AAPL
derivative counterparties (COMP) --negatively_impacts--> AAPL
```

Q2 — Answer (partially incorrect):

Based on the GRAPH FACTS, Apple discloses the following financial market conditions as negatively impacting its financial metrics: 1. Channel partner (COMP) --negativelyimpacts--> AAPL 2. Price reduction (CONCEPT) --negativelyimpacts--> AAPL (mentioned twice)

These conditions are mentioned as having a negative impact on Apple's financial metrics, such as production ramp and stock price. However, the passages do not provide specific details on the financial market conditions.

- **Q1: Succeeded**, matching ground truth exactly (SEC, DOJ, eBooks litigation).
- **Q2: Partially succeeded**. The graph query *did* correctly retrieve the key fact (`liquidity` (`FIN_MARKET`) -[`negatively_impacts`]-> `AAPL`) — solving Tier 1's retrieval failure. However, the LLM did not prioritize this fact when generating its answer, instead picking less relevant graph facts (e.g., "channel partner," "price reduction") that appeared earlier in the list, despite "liquidity" being the only fact typed as `FIN_MARKET` — the entity type that directly matches the question's own wording ("financial market conditions").
- **New finding**: this exposes a second, distinct failure mode — a **retrieval-vs-reasoning gap**. Even when the correct fact is successfully retrieved, the LLM may fail to recognize its relevance without additional structural cues. Fixed by making entity type more prominent in the prompt formatting (e.g., `[FIN_MARKET]` `liquidity` instead of burying the type at the end of the fact).
- **Conclusion**: Graph traversal reliably solves the retrieval problem that pure vector search cannot, but simply adding graph facts to a prompt is not sufficient on its own — how facts are presented to the LLM materially affects whether it uses them correctly. This is a genuinely useful, citable finding distinct from what either the FinReflectKG-MultiHop benchmark paper or the closest related agentic GraphRAG paper report.

Tier 3 — Agentic GraphRAG (Self-Directed Tool Routing)

- Built a LangChain agent equipped with three tools developed in earlier tiers: `VectorSearch` (semantic passage retrieval), `GraphTraversal` (structured relationship lookup), and `SymbolicComputation` (YoY growth calculation with provenance). Unlike Tier 2, where relevant relationship types were manually specified per question, the agent independently decides which tool(s) to call and with what input.

First run — vocabulary mismatch discovered:

For Q1, the agent's tool calls were:

```
graph_traversal_tool(input_str='AAPL|regulatory_bodies') -> No facts found
vector_search_tool(query='Apple net income litigation') -> Found eBooks litigation passage
graph_traversal_tool(input_str='AAPL|litigation') -> No facts found
```

Q1 — Answer (first run, partial):

The regulatory bodies named in connection with Apple's disclosed net income are the U.S. Department of Justice. The litigation that prompted this is the Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.).

- **Finding**: The agent guessed plausible-sounding relation type names (`regulatory_bodies` , `litigation`) that do not exist in the actual graph schema — the real relation type is `regulates` . It

only recovered a partial answer by falling back to vector search. This is not a model reasoning failure but a **tool-design failure**: the tool description did not expose the graph's actual relationship vocabulary, so the agent pattern-matched question wording to plausible relation names instead of using real ones. For Q2, the agent's guess (`negatively_impacts`) happened to match the real schema purely because the question's wording ("negatively impacting") closely mirrored the actual relation type name — a coincidence, not evidence of correct reasoning.

Fix applied: the `GraphTraversal` tool's description was updated to explicitly list all valid relation type strings present in the graph (e.g., `regulates`, `subject_to`, `negatively_impacts`, `impacted_by`, etc.), along with short usage hints (e.g., "use `regulates` for regulatory bodies/oversight").

Second run — after vocabulary fix:

For Q1, the agent's tool calls were:

```
graph_traversal_tool(input_str='AAPL|regulates')
-> [ORG_REG] u.s. department of justice --regulates--> AAPL
    [ORG_REG] u.s. district court for the southern district of new york --regulates--> AAPL
    [ORG_REG] sec --regulates--> AAPL
vector_search_tool(query='Apple net income litigation')
-> Apple eBooks Antitrust Litigation passage
```

Q1 — Answer (after fix, correct):

The regulatory bodies named in connection with Apple's disclosed net income are the U.S. Department of Justice and the U.S. District Court for the Southern District of New York, as well as the SEC. The litigation that prompted this is the Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.).

This now fully matches ground truth (all three regulatory bodies plus the correct litigation), achieved entirely through the agent's own tool selection — no manual relation-type specification was provided, unlike in Tier 2.

For Q2, the agent's tool call was:

```
graph_traversal_tool(input_str='AAPL|negatively_impacts')
-> [COMP] channel partner --negatively_impacts--> AAPL
    [CONCEPT] price reduction --negatively_impacts--> AAPL (x2)
    [FIN_MARKET] liquidity --negatively_impacts--> AAPL
```

Q2 — Answer (after fix, still partial):

Apple discloses the following financial market conditions as negatively impacting its financial metrics: 1. Price reduction 2. Liquidity

These conditions are disclosed in Apple's SEC 10-K filings as having a negative impact on the company's financial performance.

- **Finding:** Retrieval is correct — `liquidity` (the `FIN_MARKET`-typed fact) is present in the tool's raw output — but the final synthesized answer still does not fully prioritize by entity type (credit market and fixed income market, seen in earlier Tier 2 exploration, are absent from this answer). This confirms the **retrieval-vs-reasoning gap** first observed in Tier 2 persists even under fully self-directed agentic routing, making it a consistent, cross-tier finding rather than a one-off issue.
- **Also found:** a follow-up run of the graph query without a fixed sort order returned a *different* subset of facts each time (Neo4j does not guarantee result order without an explicit `ORDER BY` when using `LIMIT`), meaning some inconsistency across runs was partly due to non-deterministic retrieval rather than reasoning alone. Fixed by adding explicit ordering to the Cypher query, ensuring reproducible results across runs.
- **Overall Tier 3 conclusion:** Agentic tool routing works, but its correctness depends heavily on how well tool descriptions expose the underlying data's actual structure and vocabulary — a plausible-sounding tool interface is not sufficient if it does not match the real schema. Additionally, the retrieval-vs-reasoning gap identified in Tier 2 is not resolved by adding agentic routing alone, indicating it is a distinct problem (answer synthesis) separate from retrieval (finding the right facts), and will require targeted prompt or ranking improvements to fully resolve.

7. Remaining Work

Completed

- Data download, filtering, graph loading, indexing
- Graph pattern exploration (2-hop, 3-hop)
- Sample question set construction
- Tier 0 (no retrieval) baseline
- Tier 1 (vector RAG) — including diagnosing and fixing an embedding model quality issue
- Tier 2 (graph-enriched RAG) — including diagnosing a retrieval-vs-reasoning gap
- Symbolic computation tool — including diagnosing and partially fixing a numeric extraction bug, and documenting a multi-table disambiguation limitation
- Tier 3 (agentic GraphRAG) — including diagnosing and fixing a tool-vocabulary mismatch, and confirming the retrieval-vs-reasoning gap persists under self-directed routing

Remaining

Expand Evaluation Set - Expand from 2 to 8–10 sample questions using the same graph-pattern-mining approach, covering a wider mix of 2-hop and 3-hop, regulatory, market-risk, and metric-based patterns. - Include at least one question specifically requiring the symbolic computation tool, to test it within the full agent (not in isolation).

Full Tier Comparison + Consolidation - Run all four tiers (0–3) against the expanded question set and record accuracy, latency, and provenance-traceability for each. - Consolidate results into a comparison table for the mid-phase review. - Begin addressing the retrieval-vs-reasoning gap directly (e.g., explicit ranking/ filtering of graph facts by entity-type relevance before passing to the LLM) as a targeted improvement, since it has now been confirmed as a consistent, cross-tier issue.

This report reflects progress and near-term plans only, not the full project scope.